

ENTERPRISE MILESTONE SPECIFICATION

Module 8: Hardware Integration, QR Handheld Devices, ESC/POS Printers & 3-Tier Server Setup

Physical computing infrastructure blueprint, peripheral driver daemon gateways, POS receipt & barcode printing pipelines, and on-premises high-availability 3-tier server virtualization for Popular Auto Spare & A/C Parts Co. W.L.L.

MODULE IDENTIFIER

SL-POP-ERP-MS-008

BA REFERENCE DOCUMENT

ARCH-001 v1.0 (04-July-2026)

TOTAL ENGINEERING EFFORT

3 Working Weeks (0.75 Months)

FIXED PROJECT COST

BD 1,142.727

TARGET ARCHITECTURE

On-Premise 3-Tier HA Cluster & Peripheral DAEMONS

DOCUMENT STATUS

FINAL PROPOSAL

The **Hardware Integration & Infrastructure Platform** (Module 8) establishes the robust physical computing backbone, peripheral hardware driver gateways, thermal receipt printing pipelines, handheld Android PDA scanner synchronization daemons, and high-availability on-premises 3-tier server infrastructure for Popular Auto Spare & A/C Parts Co. W.L.L. It connects all physical retail counters, warehouse barcode terminals, label printers, and biometric attendance clocks directly to the centralized ERP core.



Core Objective & Business Purpose

To architect, deploy, and harden an enterprise-grade on-premises 3-tier virtualized server cluster, automated database replication, low-latency peripheral DAEMON gateways for Android PDA scanners, ESC/POS thermal receipt printers, Zebra industrial barcode printers, and multi-branch biometric time-clocks.

MASTER ERP TRANSFORMATION ARCHITECTURE (9 CORE PROCESS MODULES)

CURRENT SCOPE: MODULE 08
ACTIVE

01 PCode Gen & Item Master

02 Vendor & Purchase Flow

03 Stock Verification & Location

04 Sales & POS Checkout

05 Finance, Tax & VAT Accounting

06 Administration & Security

07 HRMS & Biometric Payroll

08 Hardware & Barcode Infrastructure



3-Tier Virtualized Infrastructure

On-premise Linux cluster, Nginx reverse proxy, MariaDB master-slave replication, and daily automated SAN backups.

3-Tier HA

MariaDB Cluster

SAN Backups



Barcode & Handheld Integration

Android PDA scanner background daemons for real-time stock counting, goods receipt, and dispatch verification.

PDA Daemon

QR Scanners

Live Sync



POS Thermal & Label Printing

Direct network ESC/POS thermal receipt printers and Zebra ZPL industrial barcode shelf & item label generators.

ESC/POS

Zebra ZPL

Receipts

Derived directly from **ARCH-001: Technical Architecture & Hardware Infrastructure Blueprint**, the implementation scope is structured across 5 core technical pillars:

🔧 2.1 On-Premises 3-Tier Server Virtualization & Database Clustering

- **Proxmox VE / KVM Virtualization:** Deployment of isolated Virtual Machines for Web/App layer, Database layer, and Background DAEMON services.
- **Database High Availability:** MariaDB replication cluster with automated failover and daily point-in-time recovery (PITR) snapshots.
- **Nginx Reverse Proxy & Load Balancer:** SSL termination, HTTP/2 compression, request throttling, and DDoS protection filters.
- **Automated Local & Cloud Backup Pipeline:** AES-256 encrypted night-time database dumps and transaction log sync to remote off-site storage.

📱 2.2 Android PDA Handheld Barcode / QR Scanner Integration (DAEMONS)

- **Handheld Hardware Compatibility:** Integration with Honeywell, Zebra, and Urovo industrial Android rugged PDA barcode scanners.
- **Low-Latency REST/WebSocket DAEMON:** Sub-second verification of scanned PCodes, bin locations, and purchase order items.
- **Offline Buffer Mode:** Local SQLite caching on handhelds ensuring warehouse picking and stock taking proceed during Wi-Fi blind spots.
- **Handheld Device Management:** Device authorization tokens, remote configuration push, and scanner battery telemetry monitoring.

2.3 ESC/POS POS Thermal Receipt & Zebra Industrial Barcode Label Printing

- **Direct Network ESC/POS Driver:** Raw socket printing to Epson/Bixolon 80mm POS thermal receipt printers bypassing browser print dialogs.
- **Zebra Programming Language (ZPL) Engine:** High-speed dynamic generation of 2D QR and Code-128 barcode shelf tags and item labels.
- **Multi-Terminal Print Spooler:** Centralized queue management handling concurrent receipt prints across all checkout counters without collision.
- **VAT-Compliant Invoice Layouts:** Dynamic printing of NBR-compliant simplified VAT tax invoice receipts with embedded cryptographic QR codes.

2.4 Multi-Branch Biometric Attendance Device Gateway (ZKTeco Protocol)

- **ZKTeco Standalone Push Protocol:** Gateway service capturing live biometric clock-in / clock-out punches from branch attendance terminals.
- **Device User Synchronization:** Automated two-way enrollment push of staff biometric IDs, names, and card numbers to all branch terminals.
- **Heartbeat Health Monitor:** Real-time status ping tracking online/offline connectivity of biometric clocks across all retail branches.

2.5 Network Security, SSL/TLS Encryption & Automated SAN Backup Vault

- **End-to-End SSL/TLS Encryption:** HTTPS certificates deployed across all local intranet endpoints and mobile device endpoints.
- **VLAN Network Segmentation:** Segregation of POS terminals, server cluster, Wi-Fi handhelds, and guest networks via managed switches.
- **Automated SAN Backup Vault:** Nightly differential and weekly full backup archives with automatic 30-day retention pruning.

In strict accordance with the **SaNDS Lab Dedicated Engineering Rate Card**, the resource allocation and cost breakdown for the 3-week (0.75 months / 15 working days) delivery of Module 8 is structured as follows, covering the dedicated 5-person technical squad and the Executive Steering & Domain Advisory Tier (all figures in Bahraini Dinars - BHD):

ENGINEERING ROLE	MONTHLY RATE (BHD)	WEEKLY RATE (BHD)	HOURLY RATE (160H/MO)	EFFORT ALLOCATION	TOTAL COST (BHD)
Project Manager & Enterprise Solutions Architect	BD 545.455	BD 136.364	BD 3.409	0.75 Months (3 Wks)	BD 409.092
Back-End Lead Engineer (Hardware Drivers / API Gateway)	BD 227.273	BD 56.818	BD 1.420	0.75 Months (3 Wks)	BD 170.454
Front-End Lead Engineer (Device Management Console)	BD 227.273	BD 56.818	BD 1.420	0.75 Months (3 Wks)	BD 170.454
Database & Cloud Infrastructure Architect	BD 204.545	BD 51.136	BD 1.278	0.75 Months (3 Wks)	BD 153.408
QA & System Test Automation Lead Engineer	BD 159.091	BD 39.773	BD 0.994	0.75 Months (3 Wks)	BD 119.319
Executive Steering, Domain Specialist & Security Advisory Tier	BD 160.000	BD 40.000	BD 1.000	0.75 Months (3 Wks)	BD 120.000
TOTAL MODULE 8 DEDICATED SQUAD & ADVISORY INVESTMENT	BD 1,523.636 / mo	BD 380.909 / wk	BD 9.523 / hr	3 Weeks (0.75 Mo)	BD 1,142.727



Fixed Price Guarantee

Total project investment for Module 8 is fixed at **BD 1,142.727**. No hidden deployment, integration, or hourly surcharges apply within the agreed functional scope.

The implementation is organized into 3 sequential 1-week milestones (33.33% / 33.34% each), guaranteeing rapid infrastructure deployment and verified peripheral integration:

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Milestone 8.1: 3-Tier Server Virtualization, OS Hardening & Database Clustering

Week 1 (1
Wk)

Setup of on-premises Proxmox/KVM hypervisor, Linux VM provisioning, MariaDB high-availability master-slave replication, Nginx SSL reverse proxy, and local automated daily backup cron tasks.

- ✓ **Server Hypervisor & VMs:** Provisioning of dedicated Web/App, DB, and DAEMON virtual machines with resource limits.
- ✓ **Database HA Cluster:** Replication configuration, automated failover scripts, and daily backup dumps.
- ✓ **SSL & Security Hardening:** UFW firewall configuration, SSH key policies, and Nginx SSL certificates.

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Milestone 8.2: Android PDA Scanner Daemon & ESC/POS / Zebra Label Printing Engine

Week 2 (1
Wk)

Deployment of Android PDA barcode scanning backend daemon with offline buffer, raw ESC/POS network thermal receipt printing driver, and Zebra ZPL industrial label generation pipeline.

- ✓ **PDA Scanner Service:** Sub-second PCode scanning verification and warehouse picking synchronization.
- ✓ **ESC/POS Receipt Driver:** Direct network thermal printing of simplified VAT tax invoice receipts.
- ✓ **Zebra ZPL Label Engine:** Dynamic QR and Code-128 barcode shelf and part label generation.

Milestone 8.3: Biometric Device Gateway, Automated Backup Vault & Go-Live Cutover

Week 3 (1 Wk)

Setup of ZKTeco biometric attendance gateway service, automated SAN backup verification drill, end-to-end peripheral stress testing across all retail counters, and final production cutover sign-off.

- ✓ **Biometric Device Gateway:** Live punch sync daemon from multi-branch ZKTeco clocks into HRMS.
- ✓ **Backup Restoration Drill:** Verification of point-in-time recovery and automated SAN backup replication.
- ✓ **Hardware UAT Sign-Off:** Full counter printer, scanner, and server verification with stakeholder acceptance.

Milestone Payment Structure & Invoicing Schedule (3-Week Model)

Billing is structured across 3 sequential verification gates tied directly to verifiable deliverable sign-offs:

GATE	MILESTONE DELIVERABLE FOCUS	TIMELINE	% FEE	INVOICE AMOUNT (BD)	VERIFICATION CRITERIA
Gate 8.1	3-Tier Server Virtualization & Database Clustering	End of Wk 1	33.33%	BD 380.909	Dev/Staging/Prod instances online & SSL verified
Gate 8.2	PDA Scanner Daemon & ESC/POS / Zebra Label Printing	End of Wk 2	33.33%	BD 380.909	PDA PCode barcode scan & receipt print verified
Gate 8.3	Biometric Clock Sync, SAN Backup & Final Go-Live	End of Wk 3	33.34%	BD 380.909	Biometric punch push verified & backup drill approved
Total Module 8 Investment (Fixed Deliverable Contract)			100.00%	BD 1,142.727	Complete Production Delivery

This milestone schedule is governed strictly under the overarching Master Payment Governance & Service Level Framework (SL-POP-ERP-PAY-001):

Standard Working Calendar & Effort Accounting

All engineering effort calculations are structured upon a standard 5-day working week (Sunday through Thursday), 8 working hours per day (40 hours/week, 160 hours/month). Kingdom of Bahrain statutory public holidays are observed.

15-Day Verification Window & Deemed Acceptance

Upon milestone deployment notification, the Client is granted a 15-calendar-day UAT verification period. If no material defect is logged in writing within 15 days, milestone acceptance and invoice authorization occur automatically.

90-Day Production Bug-Fix Warranty

A comprehensive 90-calendar-day post-go-live warranty covers all functional discrepancies, calculation anomalies, and system defects originating from the agreed ARCH-001 specification without additional charge.

2-Year Resource Rate Protection Guarantee

Engineering resource rates defined in Section 3.0 remain legally fixed for twenty-four (24) months from contract execution, ensuring predictable costs for any post-implementation change requests.

By executing digital signatures below, all authorized representatives formally validate the functional architecture, 3-week delivery roadmap, resource allocation pricing, and invoicing governance defined in this document:

FIRST PARTY - CLIENT / END-USER

Popular Auto Spare & A/C Parts Co. W.L.L.

Awaiting Client Signature

Signatory: Authorized Executive Director

Status: Pending Digital Execution

SECOND PARTY - LEAD IMPLEMENTATION PARTNER

SaNDS Lab Software Solutions

Awaiting SaNDS Lab Signature

Signatory: Lead Solutions Architect

Status: Pending Digital Execution

THIRD PARTY – ADVISORY & ASSURANCE PARTNER

UniGlobal Business Solutions

Awaiting UniGlobal Signature

Signatory: Principal Consultant

Status: Pending Digital Execution